

UNIT 3

TOOLS AND METHODS USED IN CYBERCRIME : Introduction, Proxy Servers and Anonymizers, Phishing, Password Cracking, Keyloggers and Spywares, Virus and Worms, Trojan-horses and Backdoors, Steganography, DoS and DDoS At-tacks, SQL Injection, Buffer Overflow, Attacks on Wireless Networks. Phishing and Identity Theft: Introduction to Phishing, Identity Theft (ID Theft).

1 INTRODUCTION

Cybercrime is not random.

Attackers use **tools + techniques + vulnerabilities** to break systems.

Cybercrime tools can be:

1. Software-based (malware, SQL injection tools)
2. Network-based (sniffers, proxy servers)
3. Human-based (phishing)
4. Code-based exploits (buffer overflow)

Engineering view:

Cybercrime = Exploiting weakness in:

- Operating System
- Network
- Application
- Human psychology

2 PROXY SERVERS AND ANONYMIZERS

2.1 What is Proxy Server?

A proxy server acts as an intermediate server between user and internet.

Normal Connection:

User → Website

Proxy Connection:

User → Proxy → Website

Website sees proxy IP, not user IP.

2.2 Why Attackers Use Proxy?

- Hide real identity

- Avoid tracing
- Bypass restrictions

2.3 Types of Proxy

1. HTTP Proxy
2. SOCKS Proxy
3. Transparent Proxy
4. Anonymous Proxy

2.4 Anonymizers

Anonymizers hide identity completely.

Examples:

- VPN
- TOR network

Engineering example:

If hacker launches DDoS directly → traceable.

If using proxy chain → difficult to track.

3 PHISHING

3.1 What is Phishing?

Phishing is a social engineering attack where attacker tricks user to reveal:

- Password
- OTP
- Credit card details

3.2 How Phishing Works?

1. Fake email sent.
2. User clicks link.
3. Fake login page appears.
4. User enters credentials.
5. Data sent to attacker.

3.3 Types of Phishing

1. Email Phishing
2. Spear Phishing
3. Whaling

4. Smishing
5. Vishing

3.4 Technical Side

Attackers create:

- Clone website
- Similar domain name

Example:

realbank.com rea1bank.com (1 instead of l)

4 PASSWORD CRACKING

Password cracking = Attempt to recover passwords.

4.1 Methods of Password Cracking

1. Brute Force Attack

Try all combinations.

2. Dictionary Attack

Try common passwords.

3. Rainbow Table Attack

Use precomputed hash values.

4. Credential Stuffing

Using leaked passwords on multiple sites.

Engineering Example:

If password is "123456" → brute force easy. If hashed with salt → harder to crack.

5 KEYLOGGERS AND SPYWARE

5.1 Keylogger

Software that records keystrokes.

Captures:

- Password
- Card details
- Messages

5.2 Types of Keyloggers

1. Software keylogger
2. Hardware keylogger (USB device)

5.3 Spyware

Spyware secretly monitors user activity.

Collects:

- Browsing history
- Screenshots
- Microphone audio

VIRUS AND WORMS

6.1 Virus

Virus attaches to legitimate file and spreads when file runs.

Needs human action.

Example:

Infected .exe file.

6.2 Worm

Self-replicating malware.

Spreads automatically via network.

Example:

Network worm infecting multiple systems.

Difference (Important Concept)

Virus needs host file.

Worm does not.

TROJAN HORSES AND BACKDOORS

7.1 Trojan Horse

Malicious software disguised as legitimate software.

Example:

Fake cracked game.

User installs → malware installed.

7.2 Backdoor

Hidden method to access system bypassing authentication.

Used for:

- Remote control
- Data theft

Engineering Example:

Developer leaves hidden admin login → backdoor.

8 STEGANOGRAPHY

Steganography = Hiding data inside another file.

Example:

- Hide secret message inside image.
- Hide malware inside audio file.

Difference:

Encryption hides content.

Steganography hides existence.

9 DoS AND DDoS ATTACKS

9.1 DoS (Denial of Service)

Single attacker floods server.

Server crashes due to overload.

9.2 DDoS (Distributed Denial of Service)

Multiple infected systems attack together.

Uses botnet.

Example:

10,000 bots sending requests to website.

10 SQL INJECTION

SQL Injection = Attack on database via input fields.

How It Happens?

If developer writes:

```
SELECT * FROM users WHERE username = ' " + input + " '
```

If attacker inputs:

```
' OR '1'='1
```

Query becomes:

```
Always true → login bypassed.
```

Reason:

Improper input validation.

Prevention:

- Prepared statements
- Parameterized queries

1 1 BUFFER OVERFLOW

Buffer overflow happens when program writes more data into memory than allocated.

Example in C:

```
If array size is 10
```

```
Input 20 characters → overwrite memory.
```

Result:

- Crash
- Execute malicious code

Technical Impact:

Attacker injects shellcode into memory.

Prevention:

- Bounds checking
- Safe languages (Java, Python)
- Stack protection

1 2 ATTACKS ON WIRELESS NETWORKS

12.1 Evil Twin Attack

Fake WiFi network created with same name.

User connects → data stolen.

12.2 Packet Sniffing

Attacker captures network traffic.

12.3 WPA Cracking

Weak WiFi password cracked.

12.4 Rogue Access Point

Unauthorized access point inside company.

1 3 PHISHING AND IDENTITY THEFT

13.1 Introduction to Phishing (Advanced)

Phishing aims to:

- Steal credentials
- Steal banking info
- Install malware

Modern phishing includes:

- QR code phishing
- OAuth phishing
- Fake job portals

13.2 Identity Theft (ID Theft)

Identity Theft = Stealing personal information to commit fraud.

Examples:

- Aadhaar misuse
- PAN misuse
- Fake loan using someone's documents

Types of Identity Theft

1. Financial identity theft
2. Medical identity theft
3. Criminal identity theft
4. Synthetic identity theft

Process of Identity Theft

1. Data collected (phishing/malware).
2. Data used for fraud.
3. Financial loss or legal issues.

1 Virus vs Worm

Basis	Virus	Worm
Dependency	Needs host file	Independent program
Spread Method	Through infected file	Through network automatically
Human Action Needed	Yes	No
Speed of Spread	Slow	Fast
Example	Infected .exe file	Network worm

Key Point: Worm spreads automatically. Virus needs execution.

2 Virus vs Trojan Horse

Basis	Virus	Trojan Horse
Nature	Self-replicating	Does not replicate
Disguise	Attaches to file	Looks like genuine software
Spread	File execution	User installs it
Purpose	Damage or spread	Create backdoor
Example	Infected document	Fake cracked software

3 Trojan Horse vs Backdoor

Basis	Trojan Horse	Backdoor
Definition	Malicious software disguised as real	Hidden access method
Purpose	Install malicious code	Bypass authentication

Visibility	Visible as software	Hidden entry
Usage	To install malware	Remote control access
Example	Fake game	Hidden admin login

4 DoS vs DDoS

Basis	DoS	DDoS
Full Form	Denial of Service	Distributed Denial of Service
Attack Source	Single system	Multiple systems
Power	Moderate	Very strong
Detection	Easier	Harder
Example	One attacker flooding server	Botnet attack

5 Proxy Server vs VPN

Basis	Proxy Server	VPN
Encryption	Usually no	Yes
Security	Moderate	High
Identity Protection	Partial	Strong
Speed	Faster	Slightly slower
Use Case	Access blocked sites	Secure browsing

6 Phishing vs Identity Theft

Basis	Phishing	Identity Theft
Nature	Method of attack	Result of attack
Purpose	Collect credentials	Use stolen identity
Medium	Email/SMS/Web	Financial systems
Example	Fake bank website	Fake loan using Aadhaar

7 Phishing vs Spear Phishing

Basis	Phishing	Spear Phishing
Target	Mass users	Specific person
Personalization	No	Yes
Success Rate	Low	High
Example	Fake bank mail to many	Fake mail to manager

8 Brute Force vs Dictionary Attack

Basis	Brute Force	Dictionary Attack
Method	Try all combinations	Try common passwords
Speed	Slow	Faster
Efficiency	Guaranteed eventually	Not guaranteed
Example	Try aaaa to zzzz	Try 123456, password

9 Encryption vs Steganography

Basis	Encryption	Steganography
Purpose	Hide content	Hide existence
Visibility	Message visible but unreadable	Message invisible
Technique	Mathematical algorithms	Hiding inside image/audio
Example	Encrypted file	Secret message in image

10 SQL Injection vs Buffer Overflow

Basis	SQL Injection	Buffer Overflow
Target	Database	Memory
Cause	Improper input validation	No bounds checking
Language	Web apps (PHP, SQL)	Low-level (C/C++)
Result	Data theft/login bypass	Execute malicious code
Prevention	Prepared statements	Safe coding practices

1 1 Keylogger vs Spyware

Basis	Keylogger	Spyware
Function	Records keystrokes	Monitors activity
Data Captured	Passwords	Browsing, screenshots
Scope	Limited	Broader
Example	Keyboard recorder	Tracking software

1 2 Wireless Sniffing vs Evil Twin Attack

Basis	Sniffing	Evil Twin
Method	Capture packets	Fake WiFi network
Goal	Monitor traffic	Steal credentials
User Interaction	No	User connects
Risk	Data leakage	Account compromise

1 3 Authentication vs Authorization

Basis	Authentication	Authorization
Meaning	Who are you?	What can you access?
Stage	First	After authentication
Example	Login	Access admin panel